

L#27: Phasor and circuit elements

Impedance and admittance

Impedance (Z): a complex number as the ratio between phasor voltage $\bar{V}$ and phasor current $\bar{I}$ of a circuit element.

$$Z = \frac{\bar{V}}{\bar{I}} = \underline{R} + jX \quad [\Omega].$$

~~$R \neq 0$~~

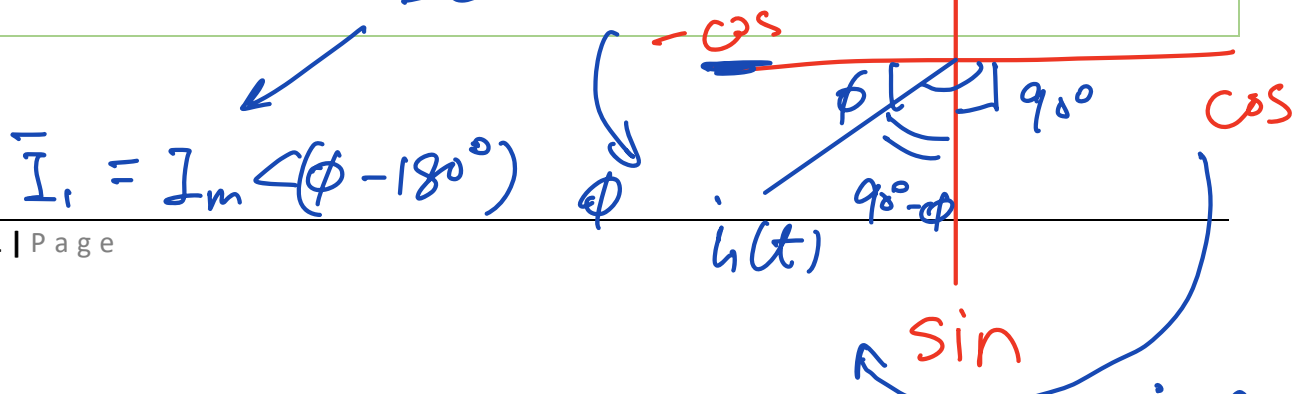
R: resistance.

X: reactance, $X > 0$, inductive reactance.
 $X < 0$, capacitive reactance.

Admittance (Y):

$$Y = \frac{1}{Z}, \quad [S].$$

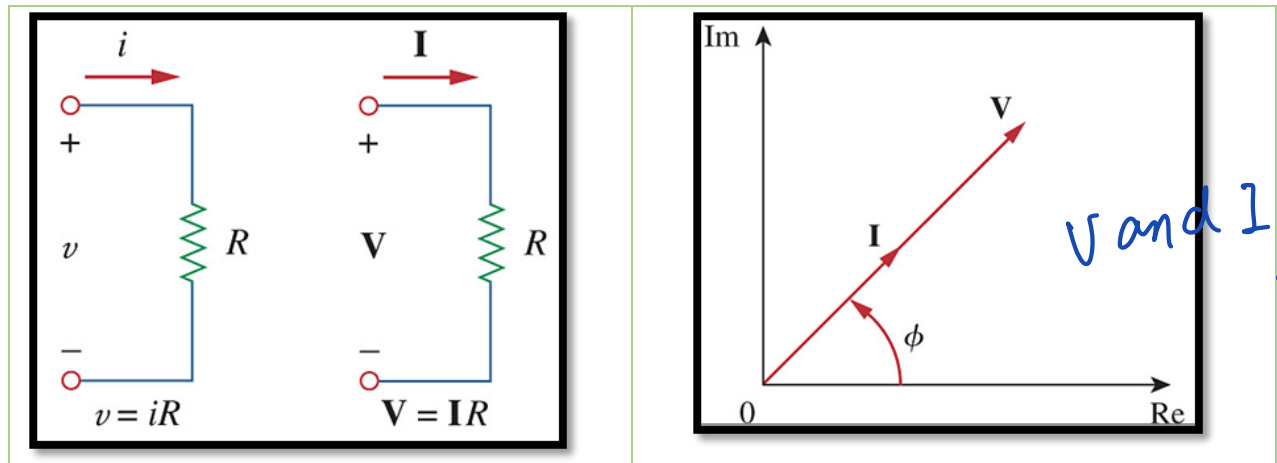
$$i_1(t) = -I_m \cos(\omega t + \phi)$$
$$= I_m \cos(\omega t + \phi - 180^\circ)$$



$$180 - \phi$$

$$\phi - 180^\circ$$

Phasor and circuit elements - resistors



$$i(t) = I_m \cos(\omega t + \phi), \quad I_m > 0.$$

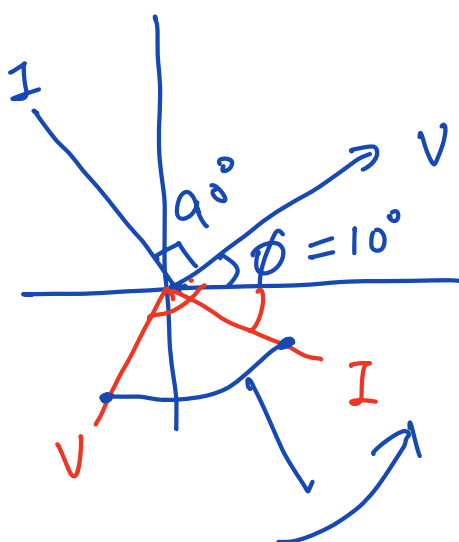
$$\bar{I} = I_m \angle \phi$$

$$v(t) = i(t) \cdot R = R \cdot I_m \cos(\omega t + \phi) = I_m R \cos(\omega t + \phi)$$

$$\bar{V} = I_m R \angle \phi.$$

$$Z_R = \frac{\bar{V}}{\bar{I}} = \frac{I_m R \angle \phi}{I_m \angle \phi} = R \angle 0^\circ = R.$$

$$1 \angle 0^\circ = 1$$



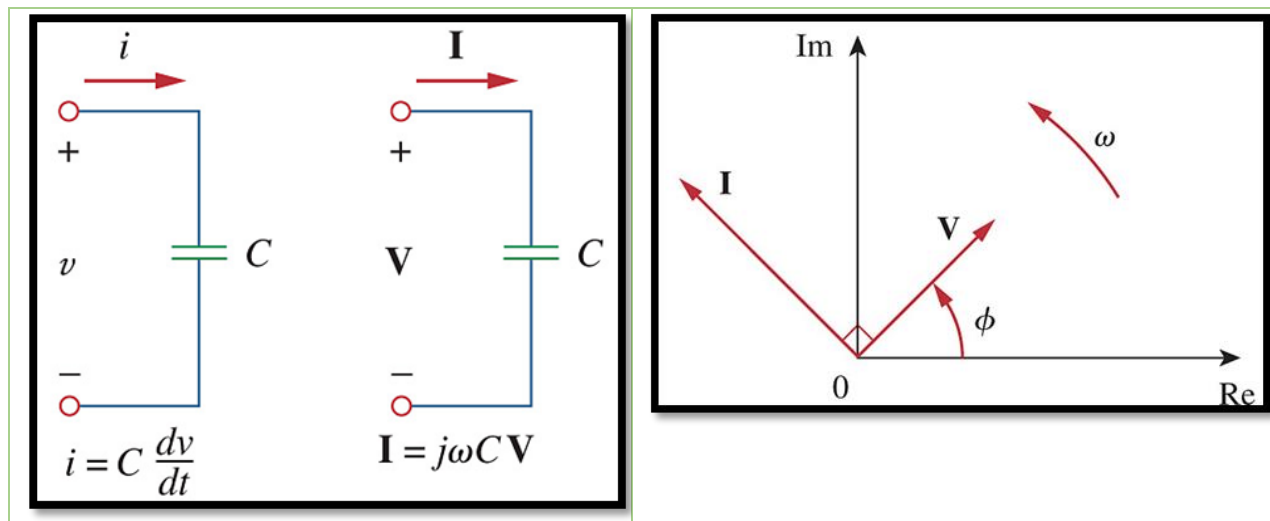
120°

$$\text{angle } V = -110^\circ.$$

$$\text{angle } I = -20^\circ.$$

i leads V .

Phasor and circuit elements - capacitors



$$v(t) = V_m \cos(\omega t + \phi), \quad \bar{V} = V_m \angle \phi$$

$$i(t) = C \frac{dv}{dt} = C \cdot v(t)' = C \cdot V_m [-\sin(\omega t + \phi)] \cdot \omega$$

$$= \omega C V_m [-\sin(\omega t + \phi)]$$

$$= \omega C V_m \cos(\omega t + \phi + 90^\circ)$$

$$\bar{I} = \omega C V_m \angle (\phi + 90^\circ)$$

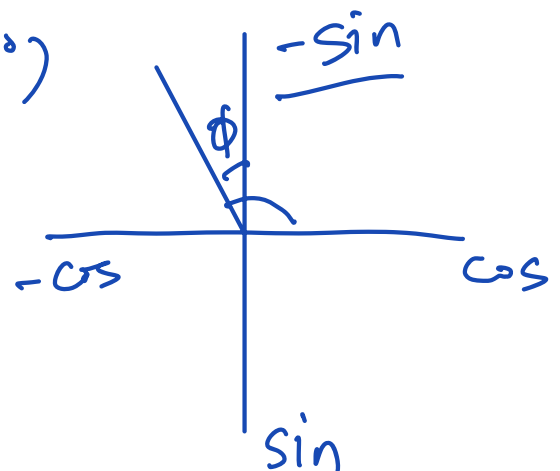
$$Z_c = \frac{\bar{V}}{\bar{I}} = \frac{V_m \angle \phi}{\omega C V_m \angle \phi + 90^\circ}$$

$$= \frac{1}{\omega C} \angle -90^\circ$$

$$= -j \frac{1}{\omega C} [\Omega]$$

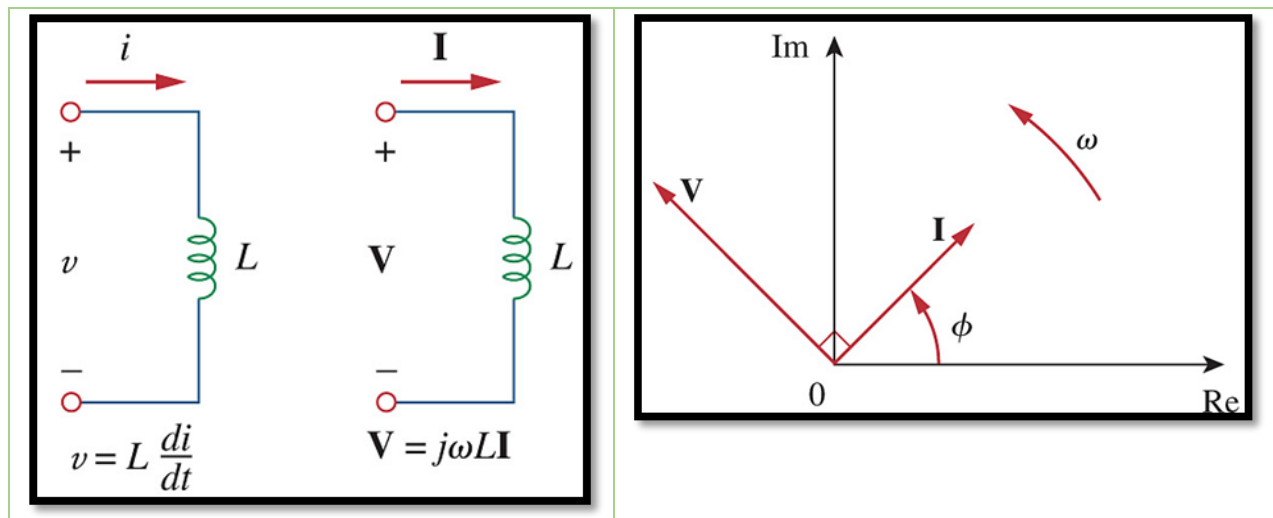
$$= \frac{1}{j\omega C} [\Omega]$$

$$Z_c = \frac{\bar{V}}{\bar{I}} = \frac{1}{j\omega C} \Rightarrow \bar{I} = j\omega C \bar{V}$$



$$1 \angle -90^\circ = -j$$

Phasor and circuit elements - inductors



$$i(t) = I_m \cos(\omega t + \phi),$$

$$v(t) = L \frac{di}{dt} = L I_m [-\sin(\omega t + \phi)] \cdot \omega.$$

$$= \omega L I_m [-\sin(\omega t + \phi)]$$

$$= \omega L I_m \cos(\omega t + \phi + 90^\circ).$$

$$\bar{V} = \omega L I_m \angle(\phi + 90^\circ), \quad \bar{I} = I_m \angle \phi$$

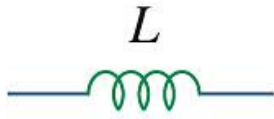
$$Z_L = \frac{\bar{V}}{\bar{I}} = \frac{\omega L I_m \angle(\phi + 90^\circ)}{I_m \angle \phi} = \omega L \angle 90^\circ = j\omega L.$$

$$\bar{V} = \bar{I} \cdot Z_L = j\omega L \cdot \bar{I}$$

Summary of voltage-current relationship		
Element	Time domain	Frequency domain
R	$v = i R$	$\bar{V}_R = \bar{I}_R \cdot R$
L	$v_L = L \frac{di}{dt}$	$\bar{V}_L = \underbrace{j\omega L}_{\text{Impedance}} \bar{I}_L$
C	$i_C = C \frac{dv}{dt}$	$\bar{I}_C = j\omega C \bar{V}_C.$

Impedances and admittances of passive elements		
Element	Impedance	Admittance
R	$Z_R = R$	$Y_R = \frac{1}{R}.$
L	$Z_L = j\omega L$	$Y_L = \frac{1}{j\omega L}$
C	$Z_C = \frac{1}{j\omega C}$	$Y_C = j\omega C.$

$$Z_L = j\omega L, \quad \omega = 2\pi f$$



Short circuit at dc



$$\text{DC}, f \rightarrow 0.$$

$$Z_L \rightarrow 0.$$

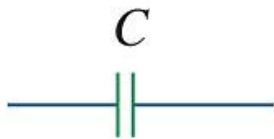
$$f \rightarrow \infty$$



Open circuit at
high frequencies

$$Z_L \rightarrow \infty$$

$$Z_C = \frac{1}{j\omega C}, \quad \omega = 2\pi f. \quad (a)$$



Open circuit at dc



$$f \rightarrow 0.$$

$$Z_C \rightarrow \infty.$$

$$f \rightarrow \infty$$



Short circuit at
high frequencies

$$Z_C \rightarrow 0.$$

(b)

Example #1:

$$\bar{V} = 10 \angle 30^\circ \text{ V.}$$

If voltage $v = 10 \cos(100t + 30^\circ)$ V is applied to a $50 \mu\text{F}$ capacitor, calculate the current through the capacitor.

$$i(t) = C \frac{dv}{dt} = 50 \times 10^{-6} \cdot 10 [-\sin(100t + 30^\circ)] \cdot 100$$

$$= 50 \times 10^{-3} [-\sin(100t + 30^\circ)]$$

$$= 50 \times 10^{-3} \cos(100t + 30^\circ + 90^\circ)$$

$$= 50 \times 10^{-3} \cos(100t + 120^\circ) \text{ A.}$$

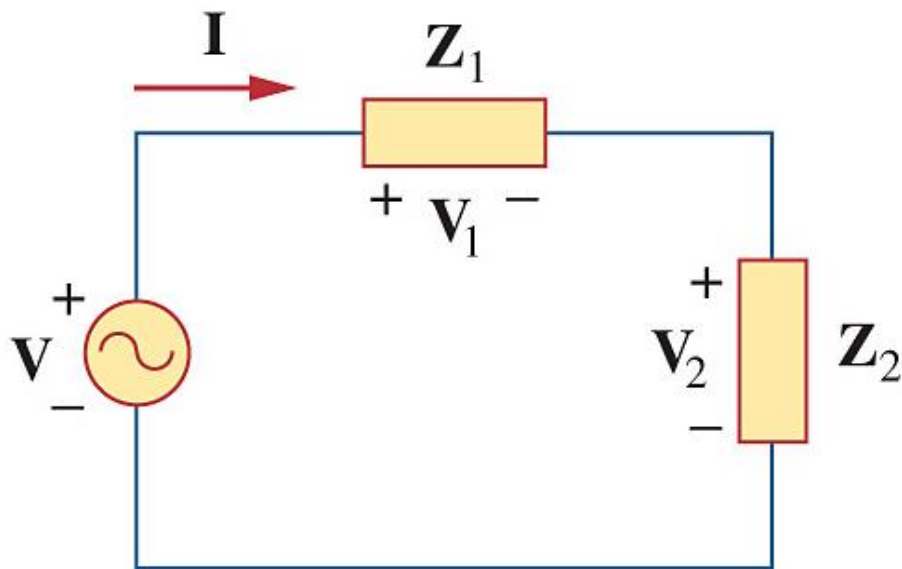
$$\bar{I} = \frac{V}{Z_C}, \quad Z_C = -j \frac{1}{\omega C} = -j \frac{1}{100 \times 50 \times 10^{-6}}$$

$$\bar{I} = \frac{10 \angle 30^\circ}{-j \frac{1}{100 \times 50 \times 10^{-6}}} = \frac{j 10 \angle 30^\circ \times 5 \times 10^{-3}}{-j^2 = 1}$$

$$= j 50 \times 10^{-3} \angle 30^\circ = 50 \times 10^{-3} \angle 120^\circ \text{ A.}$$

$$50 \times 10^{-3} \cos(100t + 120^\circ) \text{ A.}$$

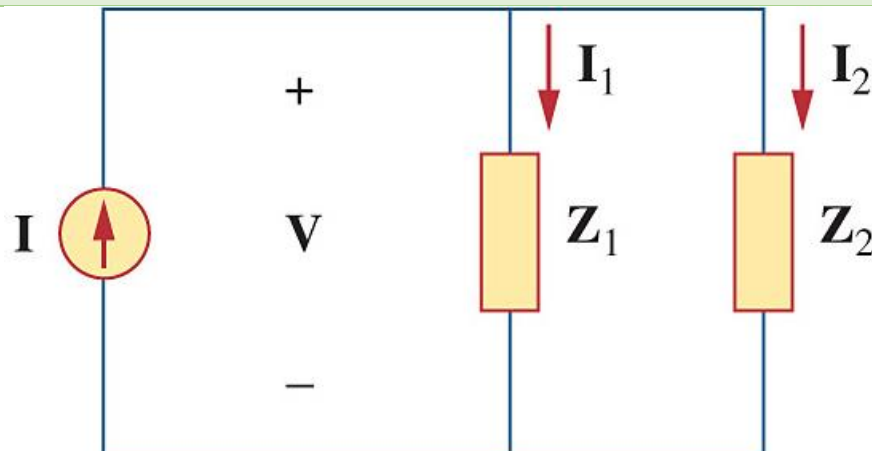
AC circuit laws and theories



(KVL, Voltage division, Z)

Voltage division applies.

$$V_1 = V \cdot \frac{Z_1}{Z_1 + Z_2}, \quad V_2 = V \cdot \frac{Z_2}{Z_1 + Z_2}$$



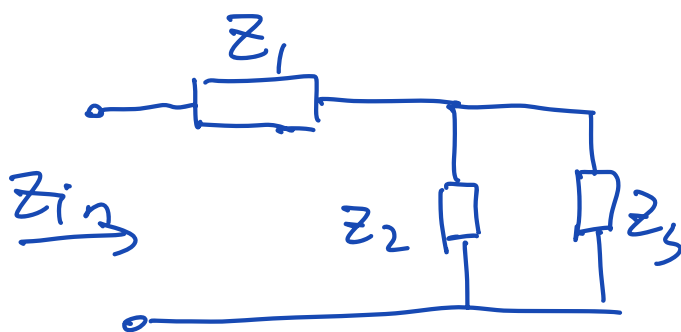
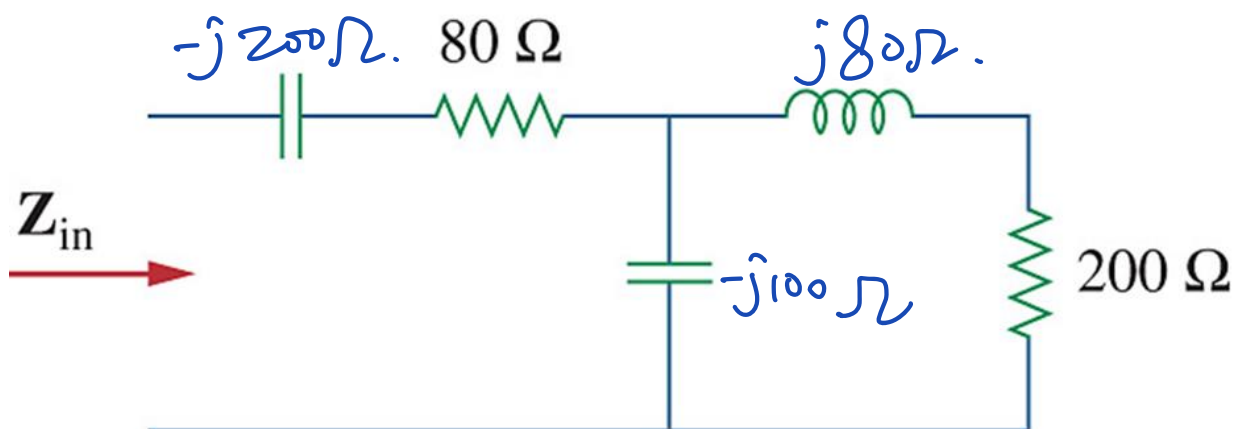
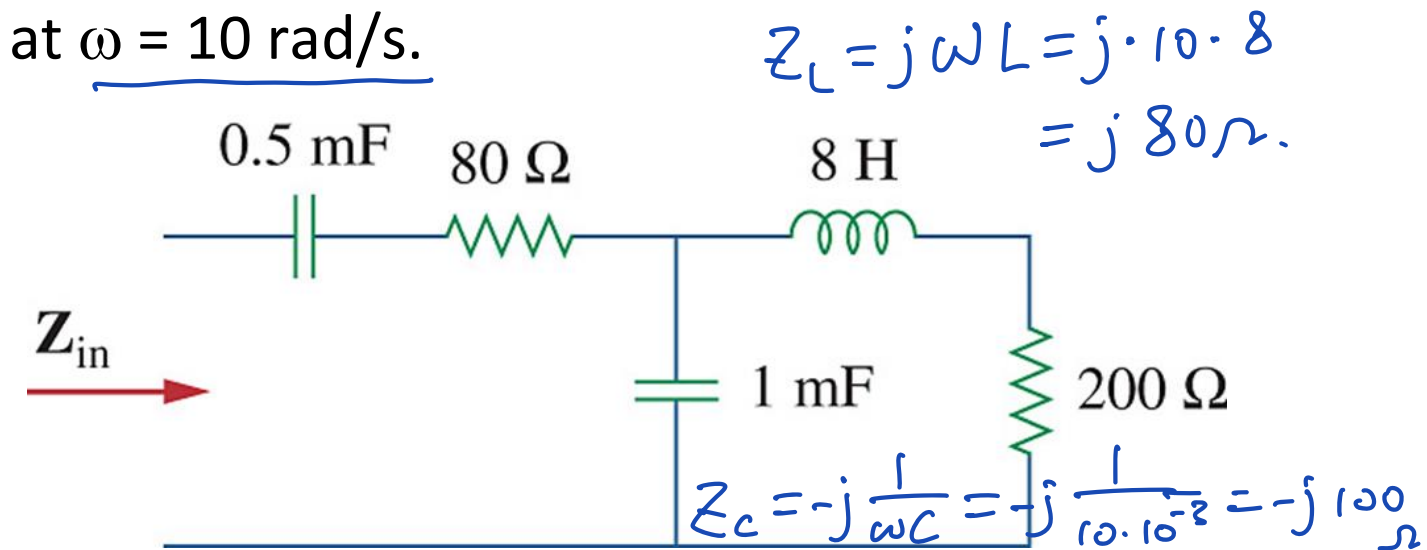
(KCL, current division, Z)

$$I_1 = I \cdot \frac{Z_2}{Z_1 + Z_2}$$

$$I_2 = I \cdot \frac{Z_1}{Z_1 + Z_2}$$

Example #2:

Determine the input impedance of the circuit at $\omega = 10 \text{ rad/s}$.



$$Z_{in} = Z_1 + Z_2 \parallel Z_3$$

$$= 80 - j200 + (-j100) \parallel (200 + j80)$$

$$= 80 - j200 + \frac{-j100(200 + j80)}{200 - j200}$$



$$= \underline{129.5 - j295} \text{ } [\Omega]$$